Project Executable Files

Date	25 September 2026
Team ID	SWTID-2026-4307
Project Name	AI Weather Wise API
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	Yes
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
      /ai-weather-wise-api |-- /app | |-- /api | | |-- /v1 | | | ├── endpoints.py | | | └── weather_ai.py | |-- /core | | ├── config.py | | └── security.py | |-- /models | | └── ml_model.pkl | └── main.py |-- /tests | └── test_api.py |-- /docs | └── swagger.json |-- Dockerfile |-- .env.example |-- README.md └-- requirements.txt
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://ai-weather-wise-api.onrender.com
Login Credentials (Demo)	Username: evaluator_demo | Password: Demo@2026Weather
Platform / Hosting Provider	Render / AWS Elastic Beanstalk
Repository Link	https://github.com/SWTID-2026-4307/AI-Weather-Wise-API
Demo Video Link	https://youtu.be/demo-ai-weather-wise

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

```
Pre-requisites: Python 3.10+, pip, virtualenv, Git Step 1: Clone the repository git clone https://github.com/SWTID-2026-4307/AI-Weather-Wise-API.git cd AI-Weather-Wise-API Step 2: Create and activate virtual environment python -m venv venv source venv/bin/activate # On Windows: venv\Scripts\activate Step 3: Install dependencies pip install -r requirements.txt Step 4: Configure environment variables cp .env.example .env Step 5: Start the application uvicorn app.main:app --reload Step 6: Open browser or API Client Navigate to http://localhost:8000/docs for Swagger UI API documentation.
```

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	API Rate limiting on external weather data provider in free tier.	Cached responses implemented using Redis; fallback mock data configured.
2	Slight latency during first cold-start request on free hosting.	Pre-warming script enabled on server initialization.
3	Historical weather analysis limited to past 5 years.	Expanded dataset support planned for v2 release.